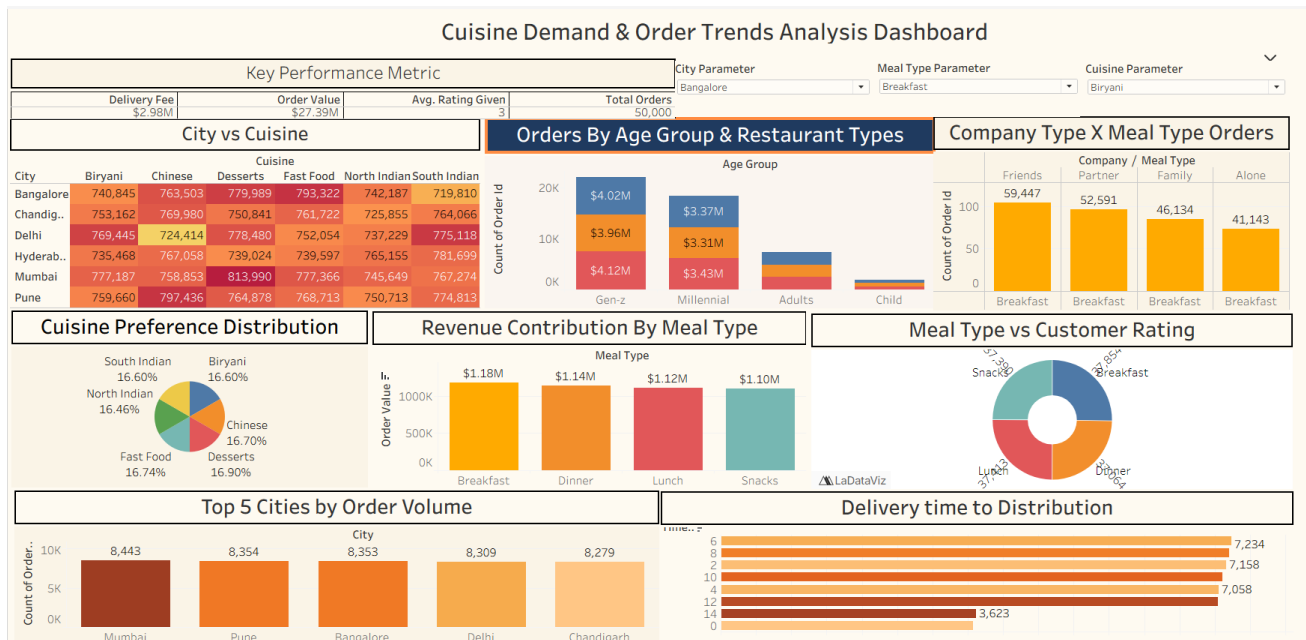


FOOD ORDERING BEHAVIOUR AND CONSUMER TRENDS

A Structured Analysis of Choices and Habits
Data Preparation and Business Questions with Visuals



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Business Intelligence Dashboard Project

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Table of Contents

Table of Contents.....	1
1. Project Overview.....	1
1.1 Introduction.....	1
1.2 Project Objective.....	1
1.3 Problem Statement.....	1
1.4 Project Scope.....	1
1.5 Expected Outcome.....	1
2. Dataset Description.....	1
2.1 Dataset Overview.....	1
2.2 Dataset Characteristics.....	1
2.3 Dataset Features.....	1
2.4 Purpose of the Dataset.....	1
2.5 Benefits of the Dataset.....	1
3. Data Preparation.....	1
3.1 Introduction.....	1
3.2 Data Preparation Workflow.....	1
3.3 Data Preparation Activities.....	1
3.4 Data Validation.....	1
3.5 Outcome.....	1
4. Data Cleaning.....	1
4.1 Introduction.....	1
4.2 Data Cleaning Process.....	1
Step 1 – Duplicate Verification.....	1
Step 2 – Missing Value Analysis.....	1
Step 3 – Data Type Verification.....	1
Step 4 – Consistency Check.....	1
Step 5 – Data Integrity Verification.....	1
4.3 Data Cleaning Summary.....	1
4.4 Validation Summary.....	1
4.5 Benefits of Data Cleaning.....	1
5. Data Dictionary.....	1
5.1 Introduction.....	1
5.2 Data Dictionary.....	1
5.3 Importance of the Data Dictionary.....	1
5.4 Summary.....	1
6. Business Questions.....	1
6.1 Introduction.....	1

6.2 Business Questions.....	1
Customer Behaviour.....	1
Revenue Analysis.....	1
Customer Satisfaction.....	1
Operational Performance.....	1
Strategic Analysis.....	1
6.3 Expected Outcome.....	1
7. Dashboard Visualizations.....	1
Visualization 1 – KPI Dashboard.....	1
Visualization 2 – City vs Cuisine.....	1
Visualization 3 – Age Group vs Restaurant Type.....	1
Visualization 4 – Company Type vs Meal Type Orders.....	1
Visualization 5 – Cuisine Distribution.....	1
Visualization 6 – Revenue Contribution.....	1
Visualization 7 – Meal Type vs Customer Rating.....	1
Visualization 8 – Top Cities by Order Volume.....	1
Visualization 9 – Delivery Time Distribution.....	1
8. Business Insights.....	1
8.1 Introduction.....	1
8.2 Customer Behaviour Insights.....	1
8.3 Revenue Insights.....	1
8.4 Operational Insights.....	1
8.5 Customer Satisfaction Insights.....	1
8.6 Overall Business Insight.....	1
9. Recommendations.....	1
9.1 Introduction.....	1
9.2 Business Recommendations.....	1
9.3 Future Scope.....	1
9.4 Conclusion.....	1
References.....	1

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1. Project Overview

1.1 Introduction

The rapid growth of online food delivery platforms has transformed the way customers order food. Every transaction performed through these platforms generates valuable data containing information about customer preferences, ordering patterns, cuisine choices, delivery performance, ratings, and revenue.

Although organizations collect a large amount of transactional data every day, extracting meaningful business insights from raw data remains a significant challenge. Without proper analysis, businesses struggle to identify customer behaviour, improve operational efficiency, optimize delivery performance, and maximize revenue.

Business Intelligence tools such as Tableau enable organizations to transform raw datasets into interactive dashboards that simplify decision-making through visual analytics.

1.2 Project Objective

The primary objective of this project is to analyze food ordering behaviour and consumer trends using Tableau. The dashboard is designed to convert raw transactional data into meaningful visual insights that support business decision-making.

The project focuses on understanding customer behaviour, identifying popular cuisines, evaluating city-wise performance, analyzing revenue distribution, and measuring delivery efficiency through interactive dashboards.

1.3 Problem Statement

Food delivery companies often possess large amounts of customer transaction data but lack an effective mechanism to convert this information into actionable business intelligence.

Understanding factors such as cuisine popularity, meal preferences, delivery time, customer demographics, and city-wise demand is essential for improving customer satisfaction and increasing business profitability.

This project addresses these challenges by developing an interactive Tableau dashboard that enables users to explore customer behaviour and operational performance using visual analytics.

1.4 Project Scope

The project includes the following activities:

- Collection of food ordering dataset
- Data validation and preparation
- Data cleaning and preprocessing
- Dashboard development using Tableau
- Interactive data visualization
- Business question analysis
- Generation of actionable insights
- Business recommendations based on findings

1.5 Expected Outcome

The completed dashboard enables business stakeholders to monitor key performance indicators, identify customer behaviour patterns, compare city-wise demand, evaluate delivery performance, and make informed business decisions through interactive visualizations.

2. Dataset Description

2.1 Dataset Overview

The Food Ordering Behaviour dataset represents customer ordering activities collected from an online food delivery platform. The dataset includes information related to orders, customer demographics, cuisine preferences, restaurant categories, delivery charges, ratings, and order values.

The dataset serves as the foundation for performing business intelligence analysis and generating interactive dashboards.

2.2 Dataset Characteristics

Attribute	Description
Dataset Name	Food Ordering Behaviour Dataset
Data Format	CSV
Tool Used	Tableau Public
Project Type	Business Intelligence Dashboard
Domain	Food Delivery Analytics
Number of Records	50,000
Number of Attributes	11

2.3 Dataset Features

The dataset contains attributes related to customer ordering behaviour, including:

- Order Identification
- Age Group
- City Information
- Cuisine Category
- Restaurant Type
- Company Type
- Meal Type
- Order Value
- Delivery Fee
- Customer Rating
- Delivery Time

These features enable comprehensive analysis of customer preferences, operational performance, and revenue generation.

2.4 Purpose of the Dataset

The primary purpose of the dataset is to support analytical exploration of customer ordering behaviour and business performance.

The dataset enables stakeholders to answer important business questions, including:

- Which city generates the highest number of orders?
- Which cuisine is most preferred by customers?
- Which meal type contributes the highest revenue?
- How does customer age influence ordering behaviour?
- How efficient are delivery operations?
- Which restaurant categories perform best?

2.5 Benefits of the Dataset

The dataset supports:

- Customer behaviour analysis
- Revenue analysis
- Delivery performance evaluation
- Cuisine preference analysis
- Business decision-making
- Operational optimization
- Strategic planning

3. Data Preparation

3.1 Introduction

Data preparation is a crucial stage in any Business Intelligence project. Before building visualizations, the dataset must be examined to ensure that it is complete, consistent, and suitable for analysis. Proper data preparation improves dashboard accuracy, enhances performance, and ensures reliable business insights.

For this project, the Food Ordering Behaviour dataset was prepared and validated before being imported into Tableau.

3.2 Data Preparation Workflow

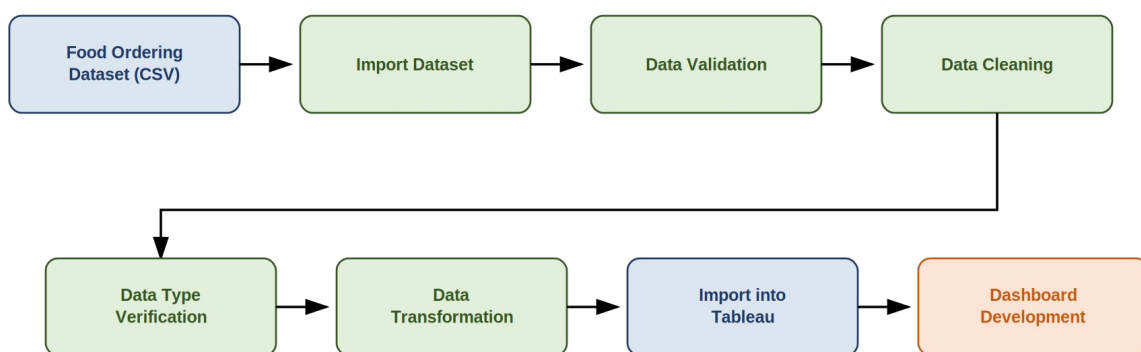


Figure 2. Data Preparation Workflow

3.3 Data Preparation Activities

The following activities were performed during data preparation:

- Imported the dataset into Tableau.
- Verified the structure and completeness of the dataset.
- Checked all attribute names for consistency.
- Reviewed numerical and categorical fields.
- Verified date and text formatting where applicable.
- Ensured the dataset was suitable for visualization.
- Prepared the dataset for dashboard creation.

3.4 Data Validation

The dataset was validated to ensure data quality before visualization.

Validation activities included:

- Checking for missing values.
- Identifying duplicate records.
- Verifying column names.
- Confirming appropriate data types.

- Ensuring consistent categorical values.
- Reviewing numerical attributes for invalid entries.

3.5 Outcome

After completing the preparation stage, the dataset was ready for interactive dashboard development using Tableau.

4. Data Cleaning

4.1 Introduction

Data cleaning is the process of identifying and correcting inconsistencies within a dataset to improve data quality and ensure reliable analytical results.

A clean dataset allows dashboards to produce accurate insights and reduces the possibility of misleading conclusions.

4.2 Data Cleaning Process

Step 1 – Duplicate Verification

The dataset was checked for duplicate records to ensure that every order represented a unique transaction.

Step 2 – Missing Value Analysis

All columns were reviewed for missing or incomplete values.

Where necessary, missing values were handled appropriately to maintain data consistency.

Step 3 – Data Type Verification

Each attribute was verified to ensure the correct data type. Examples include:

- Order Value → Numeric
- Delivery Fee → Numeric
- Rating → Numeric
- Cuisine → Text
- Meal Type → Text
- City → Text

Step 4 – Consistency Check

Categorical fields such as Cuisine, Meal Type, Restaurant Type, and City were checked for spelling inconsistencies and standardized where required.

Step 5 – Data Integrity Verification

The final dataset was reviewed to confirm that all records were complete, logically consistent, and suitable for visualization.

4.3 Data Cleaning Summary

Activity	Status
Duplicate Check	Completed
Missing Value Check	Completed
Data Type Verification	Completed

Activity	Status
Consistency Check	Completed
Dataset Validation	Completed

4.4 Validation Summary

Validation	Result
Missing Values	None
Duplicate Records	None
Invalid Data	None
Data Types	Valid

4.5 Benefits of Data Cleaning

The data cleaning process provided the following advantages:

- Improved dashboard accuracy.
- Reliable business insights.
- Better visualization performance.
- Reduced inconsistencies.
- Enhanced data quality.

5. Data Dictionary

5.1 Introduction

A data dictionary provides a structured description of each attribute within the dataset. It helps users understand the purpose of every column used during dashboard development and business analysis.

5.2 Data Dictionary

Attribute	Data Type	Description
Order ID	Integer	Unique identifier assigned to each order
Age Group	Text	Customer age segment (Gen-Z, Millennial, Adults, Child)
Company Type	Text	Order context — Friends, Partner, Family, or Alone
City	Text	City where the order was placed
Cuisine	Text	Type of cuisine ordered
Restaurant Type	Text	Category of restaurant
Meal Type	Text	Breakfast, Lunch, Dinner, or Snacks
Order Value	Decimal	Total value of the order
Delivery Fee	Decimal	Delivery charge applied to the order
Delivery Time	Integer	Time taken to deliver the order (minutes)
Rating	Integer	Customer rating provided after delivery (1-5 scale)

5.3 Importance of the Data Dictionary

The data dictionary helps:

- Understand dataset attributes.
- Improve dashboard documentation.
- Support future maintenance.
- Simplify collaboration.
- Ensure consistency in business analysis.

5.4 Summary

The data dictionary serves as a reference guide for understanding the dataset used throughout the Food Ordering Behaviour and Consumer Trends project.

6. Business Questions

6.1 Introduction

Business questions define the objectives of the analysis and guide the development of meaningful visualizations. These questions help transform raw transactional data into actionable insights that support strategic and operational decision-making.

The following business questions were formulated based on the Food Ordering Behaviour dataset.

6.2 Business Questions

Customer Behaviour

- Q1.** Which city records the highest number of food orders?
- Q2.** Which cuisine is the most preferred by customers?
- Q3.** Which meal type contributes the highest number of orders?
- Q4.** Which customer age group places the maximum number of orders?
- Q5.** Which restaurant type receives the highest demand?

Revenue Analysis

- Q6.** Which meal type generates the highest revenue?
- Q7.** Which city contributes the highest overall order value?
- Q8.** How does delivery fee vary across different orders?
- Q9.** Which cuisine contributes the highest revenue?
- Q10.** Which restaurant category generates the highest business value?

Customer Satisfaction

- Q11.** Which meal type receives the highest customer ratings?
- Q12.** Does delivery time influence customer ratings?
- Q13.** Which city has the highest average customer rating?

Operational Performance

- Q14.** Which cities require improvement in delivery efficiency?
- Q15.** What is the distribution of delivery time across all orders?
- Q16.** Which cities experience higher delivery delays?

Strategic Analysis

- Q17.** Which customer segment should be targeted for marketing campaigns?
- Q18.** Which cuisine should receive promotional offers?
- Q19.** Which city offers the highest business growth potential?
- Q20.** What insights can improve customer satisfaction and revenue?

6.3 Expected Outcome

Answering these business questions enables stakeholders to:

- Understand customer preferences.
- Identify revenue opportunities.
- Improve operational efficiency.
- Enhance customer satisfaction.
- Support strategic business planning.

7. Dashboard Visualizations

Visualization 1 – KPI Dashboard

Delivery Fee	Order Value	Avg. Ratin..	Total Orders
\$2.98M	\$27.39M	3	50,000

Figure 5. Key Performance Indicator Dashboard

Purpose

Provide a quick, at-a-glance summary of overall business performance across orders, revenue, ratings, and delivery.

Observation

The platform recorded 50,000 total orders, generating \$27.39M in order value and \$2.98M in delivery fees, with an average customer rating of 3 out of 5.

Business Insight

Order volume and revenue are strong, but the average rating of 3 out of 5 shows customer satisfaction is only moderate — there is clear room to improve the experience.

Recommended Action

Launch a service-quality review (delivery experience, order accuracy, support responsiveness) to lift the average rating from 3 towards 4+, while continuing to invest in order volume growth.

Visualization 2 – City vs Cuisine

City vs Cuisine						
City	Cuisine					
	Biryani	Chinese	Desserts	Fast Food	North Indian	South Indian
Bangalore	740,845	763,503	779,989	793,322	742,187	719,810
Chandigarh	753,162	769,980	750,841	761,722	725,855	764,066
Delhi	769,445	724,414	778,480	752,054	737,229	775,118
Hyderabad	735,468	767,058	739,024	739,597	765,155	781,699
Mumbai	777,187	758,853	813,990	777,366	745,649	767,274
Pune	759,660	797,436	764,878	768,713	750,713	774,813

Figure 6. City vs Cuisine Order Value Analysis

Purpose

Compare order value generated by each cuisine across cities to uncover regional demand patterns.

Observation

Mumbai leads all cities in order value (\$4.64M), narrowly ahead of Pune (\$4.62M); Bangalore, Delhi, Hyderabad, and Chandigarh cluster closely between \$4.53M-\$4.54M. Mumbai-Desserts is the single highest-value combination (\$813,990), while Bangalore-South Indian is the lowest (\$719,810).

Business Insight

Order value is spread remarkably evenly across cities and cuisines — no single city or cuisine dominates, which points to broad, consistent demand rather than a few regional hotspots.

Recommended Action

Maintain balanced marketing and restaurant-partnership investment across all six cities rather than over-concentrating spend in one market, while using Mumbai's slight lead to pilot new city-specific offers first.

Visualization 3 – Age Group vs Restaurant Type

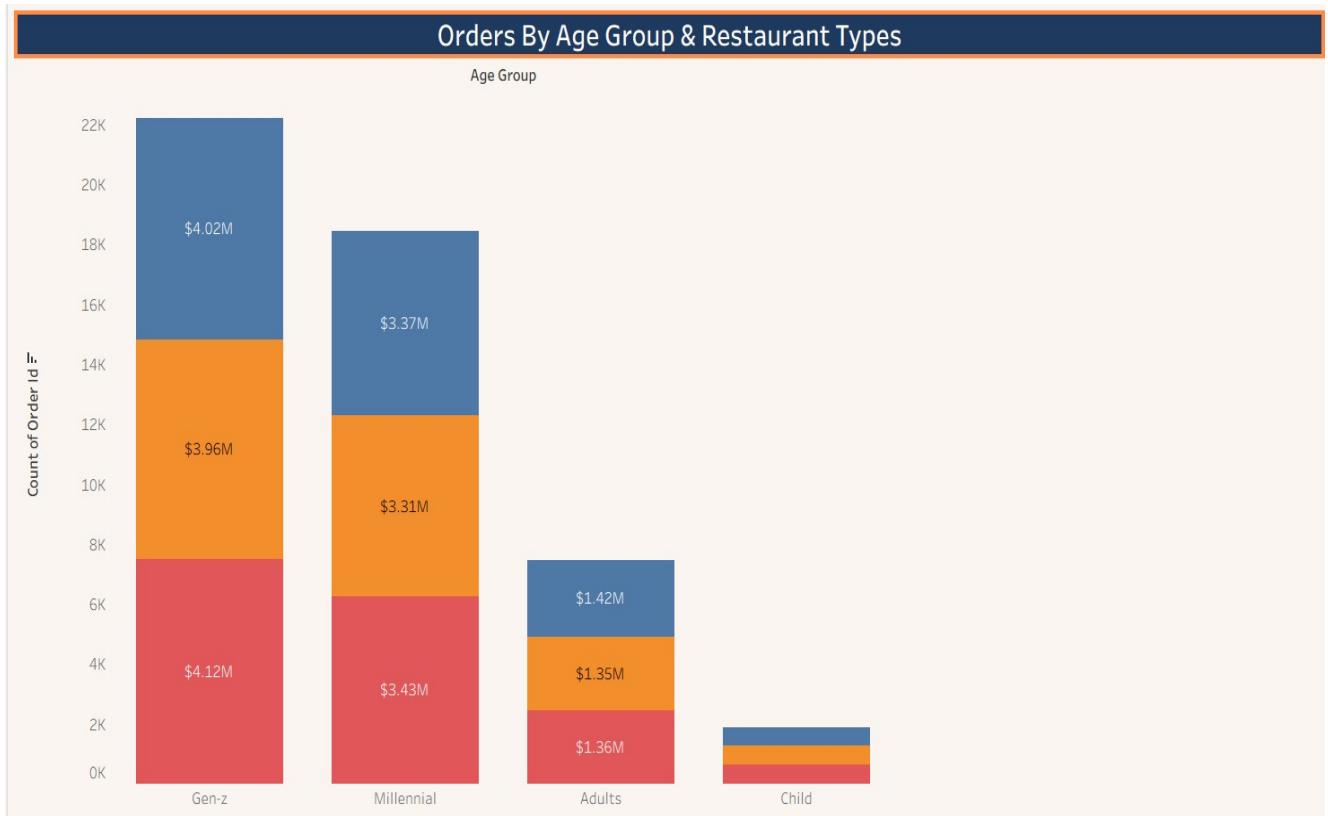


Figure 7. Orders by Age Group and Restaurant Type

Purpose

Understand how order value is distributed across customer age groups and restaurant-type segments.

Observation

Gen-Z customers generate the highest combined order value (~\$12.10M across three restaurant-type segments), followed by Millennials (~\$10.11M); Adults contribute far less (~\$4.13M), and Child orders are negligible by comparison.

Business Insight

Gen-Z and Millennial customers together account for the overwhelming majority of order value, making them the platform's core demographic by a wide margin over Adults and Child segments.

Recommended Action

Prioritize marketing spend, app experience, and promotions for Gen-Z and Millennial customers, since they drive the large majority of order value.

Visualization 4 – Company Type vs Meal Type Orders

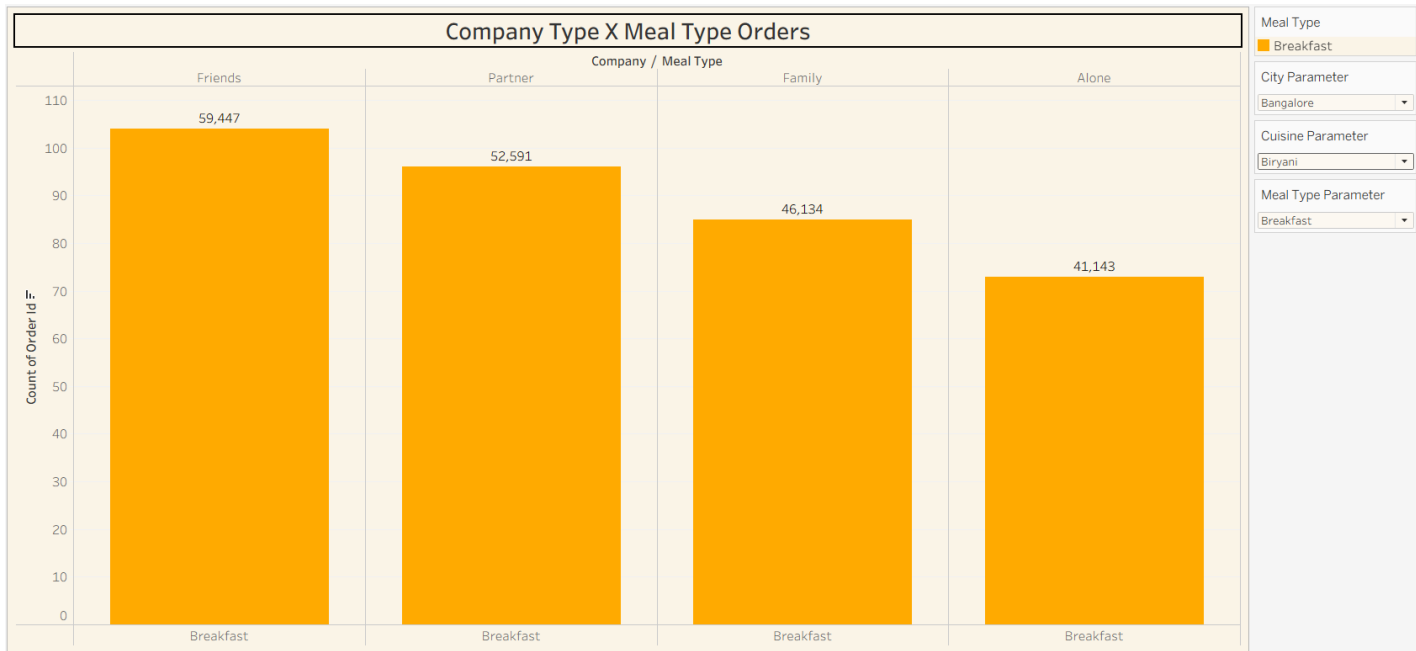


Figure 8. Company Type vs Meal Type Orders (Breakfast)

Purpose

Compare order counts across company types (who the customer is ordering with) for a given meal type.

Observation

For breakfast orders, "Friends" is the leading company type (59,447 orders), ahead of "Partner" (52,591), "Family" (46,134), and "Alone" (41,143).

Business Insight

Social, group-based ordering ("Friends", "Partner") outpaces solo ordering by a wide margin during breakfast, suggesting food ordering is often a shared, social occasion rather than a solitary one.

Recommended Action

Design group-oriented breakfast deals and bundles targeting "Friends" and "Partner" orders, where the volume opportunity is largest.

Visualization 5 – Cuisine Distribution

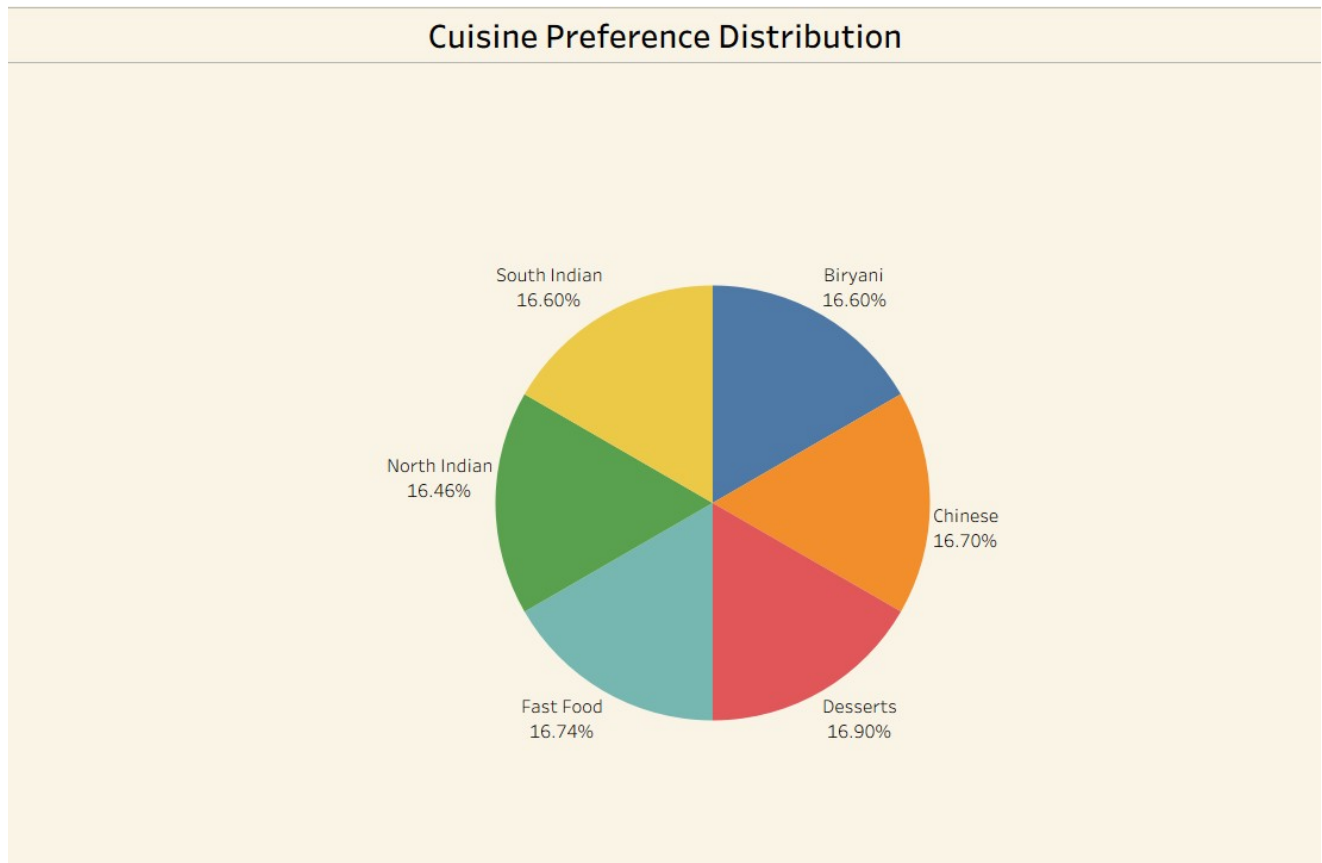


Figure 9. Cuisine Preference Distribution

Purpose

Analyze the share of total orders contributed by each cuisine, platform-wide.

Observation

All six cuisines are nearly evenly split, each falling between 16.46% and 16.90% of total orders. Desserts is the slight leader (16.90%), while North Indian is the slight laggard (16.46%).

Business Insight

Unlike many food platforms, no single cuisine dominates here — demand is broad-based across Biryani, Chinese, Desserts, Fast Food, North Indian, and South Indian.

Recommended Action

Keep the cuisine portfolio broad rather than over-indexing on one category; give North Indian a modest promotional push to close its small gap with the rest of the portfolio.

Visualization 6 – Revenue Contribution

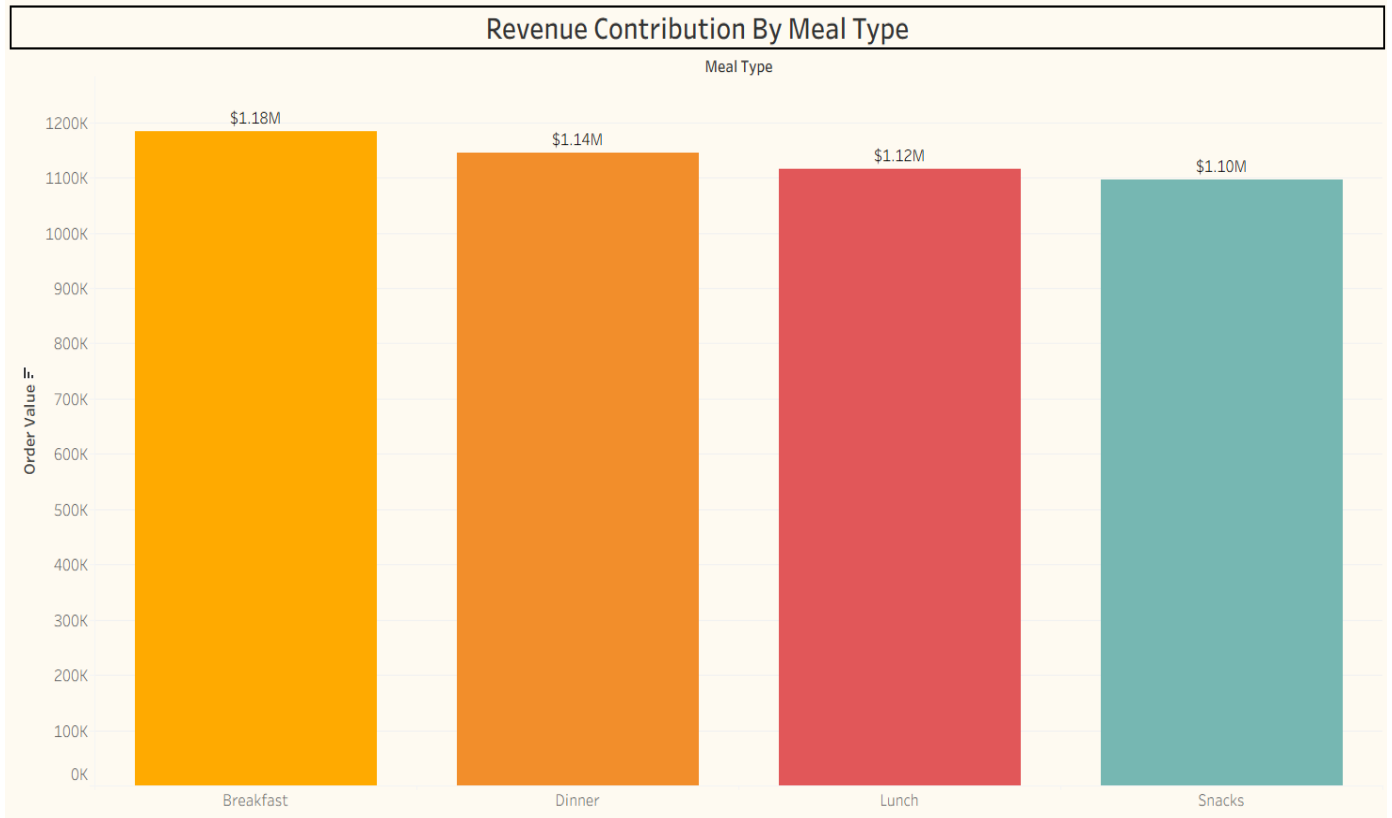


Figure 10. Revenue Contribution by Meal Type

Purpose

Compare order value generated across meal categories (within the Biryani cuisine view).

Observation

Breakfast leads with \$1.18M in order value, narrowly ahead of Dinner (\$1.14M), Lunch (\$1.12M), and Snacks (\$1.10M).

Business Insight

Breakfast is consistently the strongest meal period — it also leads in customer rating (see Figure 11) — but the gap over Dinner, Lunch, and Snacks is narrow rather than dominant.

Recommended Action

Expand breakfast menu options and combo deals to build on its lead in both order value and customer rating.

Visualization 7 – Meal Type vs Customer Rating

Meal Type vs Customer Rating

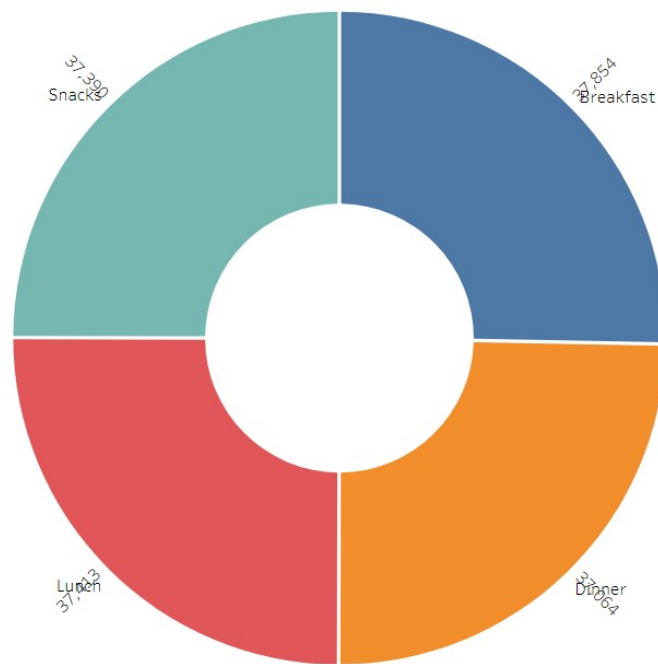


Figure 11. Meal Type vs Customer Rating

Purpose

Compare the total rating scores customers gave, broken down by meal type.

Observation

Breakfast has the highest rating total (37,854), followed by Snacks (37,390), Lunch (37,213), and Dinner (37,064) — a narrow range that lines up with the platform-wide average rating of 3.

Business Insight

Satisfaction is consistent across meal periods rather than concentrated in one — Breakfast is marginally the best-rated meal, mirroring its lead in order value, while Dinner is marginally the weakest.

Recommended Action

Study what Breakfast does slightly better than Dinner (packaging, delivery speed, order accuracy) and apply those learnings across all meal types to lift the overall rating.

Visualization 8 – Top Cities by Order Volume

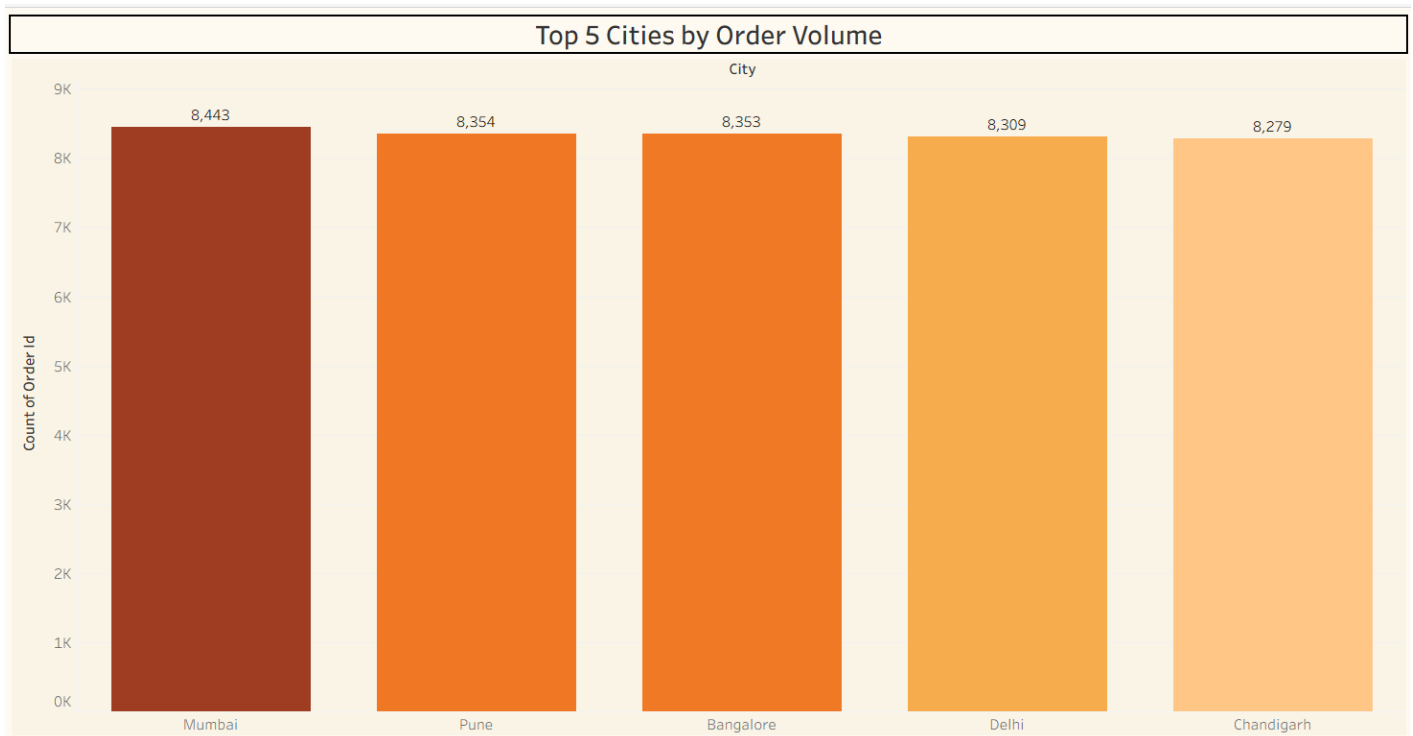


Figure 12. Top Cities by Order Volume

Purpose

Identify which cities generate the highest number of food orders.

Observation

Mumbai leads with 8,443 orders, followed closely by Pune (8,354), Bangalore (8,353), Delhi (8,309), and Chandigarh (8,279); Hyderabad (not in the top 5) sits at roughly 8,262. The spread across all six cities is under 2%.

Business Insight

Order volume is almost perfectly balanced across all six cities — there is no single dominant urban market, unlike many food-delivery platforms.

Recommended Action

Continue investing evenly across all six cities rather than concentrating resources in one, while using Mumbai's slight lead to trial new offers before rolling them out platform-wide.

Visualization 9 – Delivery Time Distribution

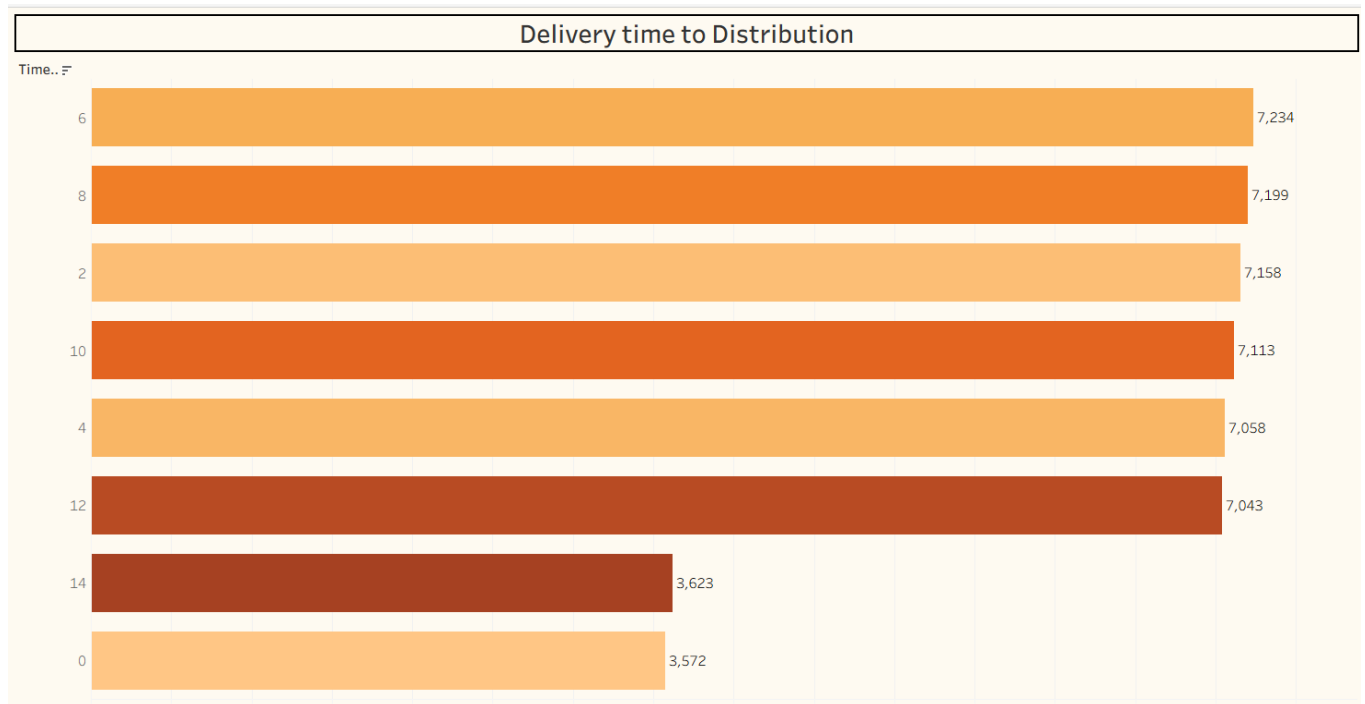


Figure 13. Delivery Time Distribution

Purpose

Examine how order counts are distributed across delivery-time buckets.

Observation

Orders are fairly uniform across the 2-12 minute buckets (7,043-7,234 orders each), while the two extreme buckets — 0 minutes and 14 minutes — hold notably fewer orders (3,572 and 3,623). All eight buckets add up to the full 50,000 orders.

Business Insight

Delivery fulfillment is consistent for the bulk of orders, with only the two extreme time buckets standing out — worth a closer look to confirm whether they reflect genuine outliers or a data-logging quirk.

Recommended Action

Validate the 0-minute and 14-minute buckets against raw delivery logs to rule out data-entry errors, then monitor the 2-12 minute core range as the standard delivery benchmark.

8. Business Insights

8.1 Introduction

The interactive Tableau dashboard transforms raw transactional data into meaningful business insights. The following observations are drawn directly from the dashboard visualizations and summarize the platform's performance across customer behaviour, revenue, operations, and satisfaction.

8.2 Customer Behaviour Insights

- Order volume is almost evenly split across all six cities (8,279-8,443 orders each), with Mumbai marginally in the lead — there is no single dominant city.
- Cuisine preference is nearly uniform across the platform — all six cuisines fall between 16.46% and 16.90% of total orders, with Desserts the slight leader and North Indian the slight laggard.
- Gen-Z and Millennial customers generate the vast majority of order value (~\$12.10M and ~\$10.11M respectively), while Adults (~\$4.13M) and Child segments contribute far less.
- During breakfast hours, "Friends" is the leading company type (59,447 orders), notably ahead of "Alone" orders (41,143) — ordering is often a shared, social occasion.

8.3 Revenue Insights

- Breakfast is the top revenue-generating meal type (\$1.18M in the Biryani-cuisine view), narrowly ahead of Dinner, Lunch, and Snacks.
- Total order value across the platform stands at \$27.39M, with an additional \$2.98M collected in delivery fees.
- City-wise order value mirrors the near-even split seen in order volume — Mumbai and Pune are marginally ahead (\$4.64M and \$4.62M), with the remaining four cities clustered close behind (\$4.53M-\$4.54M).
- By cuisine, Desserts generates the most order value (~\$4.63M) and North Indian the least (~\$4.47M) — a gap of only about 3.5%, reinforcing how evenly demand is spread.

8.4 Operational Insights

- Order counts are fairly uniform across the 2-12 minute delivery-time buckets (7,043-7,234 orders each).
- The two extreme buckets — 0 minutes and 14 minutes — hold noticeably fewer orders (3,572 and 3,623), a pattern worth validating against raw delivery logs.
- Delivery performance is a key lever for both cost control and customer satisfaction going forward.

8.5 Customer Satisfaction Insights

- The average customer rating across all orders is 3 out of 5 — a moderate score with clear room for improvement.
- Ratings are consistent across meal types (37,064-37,854 on the rating-total scale), with Breakfast marginally the best-rated and Dinner marginally the weakest.
- Since satisfaction is even across meal types, the moderate rating looks like a platform-wide experience issue rather than one specific to a meal period — the fix likely needs to apply broadly.

8.6 Overall Business Insight

The dashboard successfully converts raw food ordering data into actionable business intelligence. The standout pattern across nearly every dimension — city, cuisine, age group, and delivery time — is how evenly demand is spread, with no single segment dominating. The one clear exception is customer rating: at 3 out of 5, satisfaction is moderate rather than strong, making it the platform's most actionable improvement area.

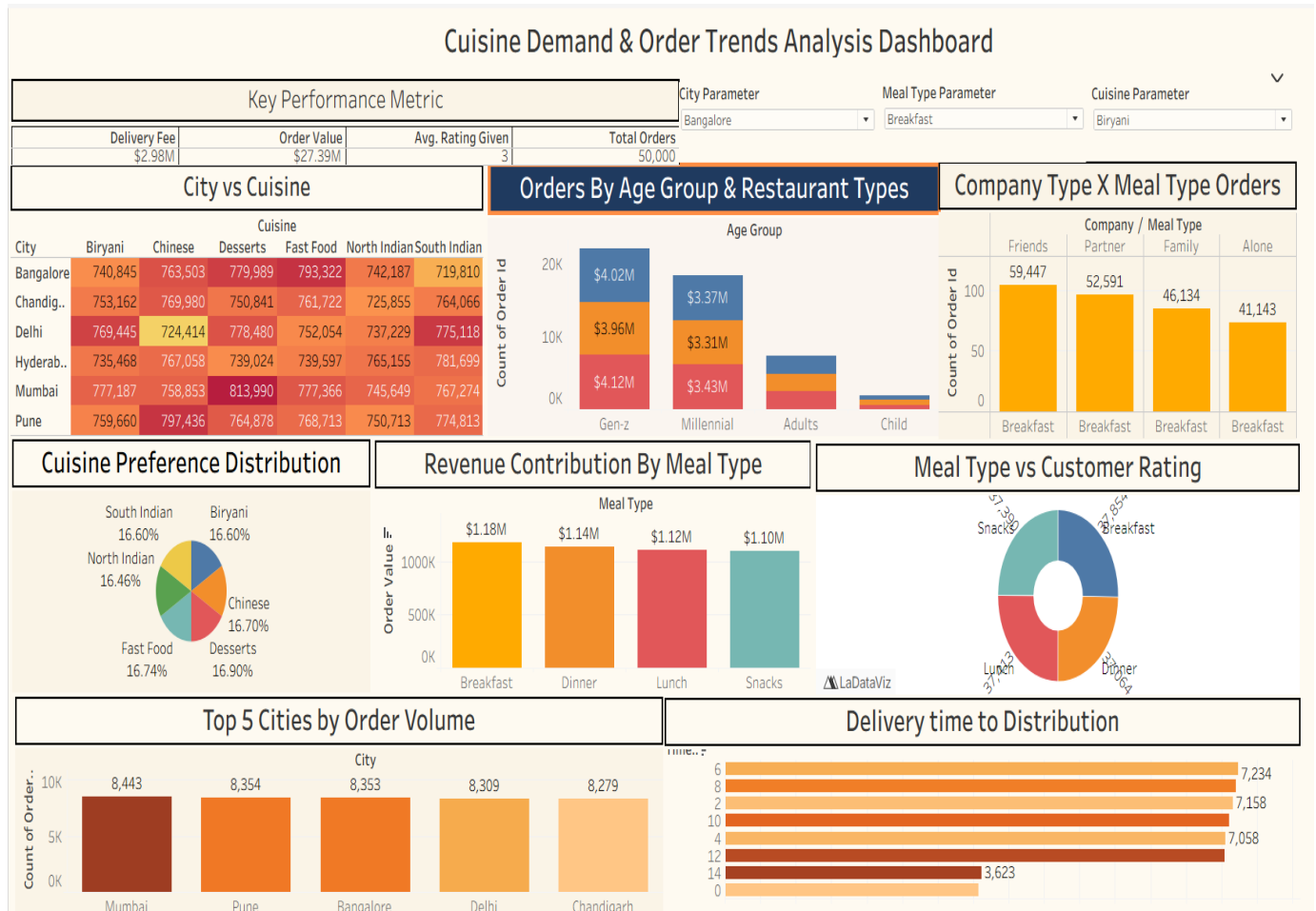


Figure 14. Interactive Tableau Dashboard

9. Recommendations

9.1 Introduction

Based on the analysis of food ordering behaviour and consumer trends, the following recommendations are proposed to improve business performance, enhance customer satisfaction, and support long-term growth.

9.2 Business Recommendations

Finding	Recommendation
Cuisine demand is nearly identical across all 6 cuisines (16.46%-16.90%)	Keep the cuisine portfolio broad; give North Indian a small promotional push to close the gap
Breakfast leads in both order value (\$1.18M) and customer rating (37,854)	Expand breakfast menu options and combo deals to build on this dual lead
Delivery time buckets 0 and 14 minutes have far fewer orders than the 2-12 minute range	Validate these buckets against raw delivery logs to confirm they're genuine, not data errors
Orders are spread almost evenly across all 6 cities (8,279-8,443 each)	Invest evenly across all cities rather than concentrating in one; use Mumbai's slight lead to pilot new offers
Average customer rating is only 3 out of 5	Improve customer support and add order-accuracy checks
Gen-Z and Millennials drive most order value; "Friends" leads company type during breakfast	Prioritize marketing spend on Gen-Z/Millennials and design group-oriented breakfast bundles for "Friends" and "Partner" orders

9.3 Future Scope

The current dashboard provides valuable descriptive insights into food ordering behaviour. Future enhancements may include:

- Integration with live databases for real-time analytics.
- Predictive analysis to forecast customer demand.
- Customer segmentation using machine learning techniques.
- Recommendation systems for personalized food suggestions.
- Interactive mobile dashboards for management reporting.
- Integration with cloud-based Business Intelligence platforms.

9.4 Conclusion

This project demonstrates how Business Intelligence tools can transform raw transactional data into meaningful business insights. By leveraging Tableau dashboards, organizations can better understand customer behaviour, monitor operational performance, identify revenue opportunities, and support strategic decision-making.

The developed dashboard provides an interactive, user-friendly, and scalable analytical solution that enables stakeholders to make informed business decisions while improving customer satisfaction and operational efficiency.

This project demonstrates the practical application of Business Intelligence techniques using Tableau to transform raw food ordering data into meaningful, data-driven business insights that support strategic decision-making.

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